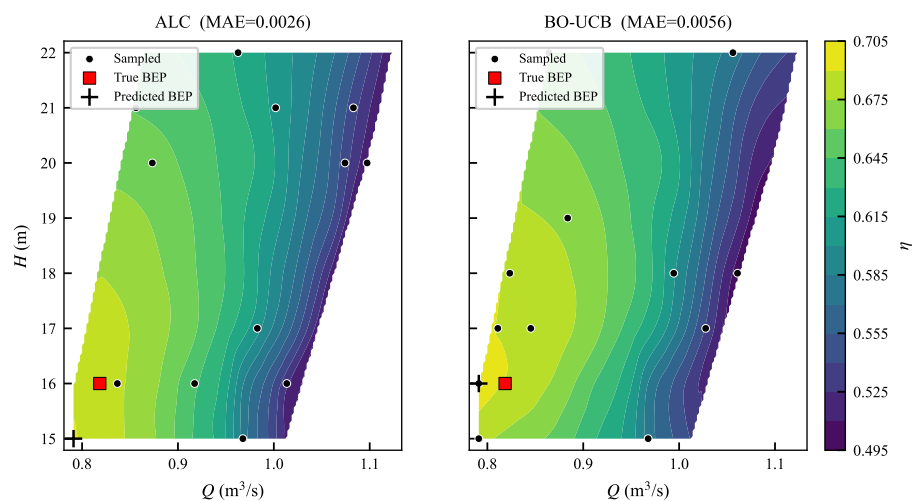


Graphical Abstract

Data-Driven Sequential Decision-Making for Small Hydro Propeller Turbine Performance Testing: Bayesian Optimization and Active Learning for Experiment Reduction

Siwakorn Suprasertchai, Kornpaphop Ruttanawijit, Yodchai Tiaple



Highlights

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- Turbine performance testing is framed as a budget-constrained sequential decision problem.
- Six sampling policies are benchmarked on 72 real propeller-turbine measurements.
- Active learning reconstructs the full hill chart to $R^2 \approx 0.996$ using only 12 of the 72 test points.
- Bayesian optimization locates the exact best efficiency point from only 14 test points.
- The two objectives select opposite winners; the sampler must match the deliverable.

Data-Driven Sequential Decision-Making for Small Hydro Propeller Turbine Performance Testing: Bayesian Optimization and Active Learning for Experiment Reduction

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Abstract

Constructing the efficiency map (hill chart) of a hydraulic turbine and locating its best efficiency point (BEP) requires many physical experiments, each costly in both time and labor. Existing approaches either couple surrogate models to CFD-based geometric design or fit models to pre-collected test data under a fixed sampling plan; the use of an adaptive policy to decide which physical experiments to run, benchmarked against classical design of experiments on real measurements, remains unaddressed. This work frames performance testing as a sequential decision problem and compares six sampling policies: Bayesian optimization with an upper-confidence-bound acquisition (BO-UCB), two Gaussian-process active-learning policies (ALM and ALC), and random, grid, and Latin hypercube sampling. Sweeping the experimental budget from 6 to 30 points over 20 random seeds, we evaluate every policy against a 72-point baseline measured on a 137 kW small-hydro propeller turbine. For reconstructing the full hill chart the active learners dominate: ALC reaches $R^2 = 0.996$ and $\text{MAPE} = 0.35\%$ from only 12 experiments (an 83.3% reduction), whereas BO-UCB is the weakest at reconstruction ($R^2 = 0.921$). For locating the BEP the ranking inverts: BO-UCB identifies the exact BEP node ($Q = 0.8185 \text{ m}^3 \text{ s}^{-1}$, $H = 16.00 \text{ m}$, $\eta^* = 0.6973$) on every seed from

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14 experiments (an 81% reduction), while all other policies plateau near $\Delta\eta \approx 10^{-2}$. Every separation is confirmed by paired Wilcoxon signed-rank tests. These results show that the acquisition function must match the deliverable: global fit is the wrong success metric for BEP-directed testing, and BEP-directed sampling is the wrong tool for building the map.

Keywords: Bayesian optimization, sequential experimental design, small hydropower, propeller turbine, surrogate modeling, active learning

1. Introduction

1.1. Background and motivation

Small hydropower remains one of the most mature and dispatchable sources of renewable electricity, and propeller-type reaction turbines are widely deployed in run-of-river and low-head installations where the available head is modest but the flow is comparatively large. The operational value of such a machine is governed by its efficiency characteristic over the full envelope of discharge and head at which it is expected to run. This characteristic is summarized by the efficiency map, or hill chart, which expresses turbine efficiency as a function of the principal operating variables and locates the best efficiency point (BEP)—the operating condition at which the machine converts the available hydraulic power into shaft power most effectively. Accurate knowledge of the hill chart and of the BEP underpins site matching, control-system tuning, performance guarantees, and condition monitoring throughout the service life of the unit.

Constructing a reliable hill chart from physical measurements is expensive. Each test point requires the rig to be brought to a stable operating condition, the instrumentation to settle, and repeated readings to be averaged, all under controlled procedures. A densely sampled map of the kind used as ground truth in the present study—comprising several dozen individual operating points across the discharge–head plane—represents a substantial commitment of test-rig time, energy, and labor. In commercial and research settings alike, this cost places a practical ceiling on how finely the operating envelope can be characterized, and it motivates a question that is fundamentally one of resource allocation: given a limited budget of physical experiments, which operating points should be measured so that the quantities of engineering interest are recovered as accurately as possible? Propeller and other reaction turbines are the dominant choice in the low- and

29 ultra-low-head range, where compact run-of-river installations are increas-
 30 ingly important for distributed renewable generation [1], and small-hydro
 31 units of this class have been deployed and characterized in run-of-river set-
 32 tings in Thailand [2]. Standardized model- and prototype-acceptance testing
 33 of such machines follows international codes [3], whose controlled procedures
 34 are precisely what make each measured operating point costly.

35 *1.2. The decision problem in turbine performance testing*

36 The question above is best framed not as a curve-fitting problem but as
 37 a sequential decision-making problem under expensive evaluation. At each
 38 stage of a test campaign, the engineer has already observed efficiency at a
 39 set of operating points and must decide where to run the next experiment.
 40 The decision is made under uncertainty, because the efficiency at untested
 41 conditions is unknown, and it is made under a budget constraint, because
 42 every additional experiment consumes scarce resources. A good decision pol-
 43 icy allocates the limited budget to the points that are most informative for
 44 the goal at hand, rather than spreading effort uniformly across a region of
 45 the operating envelope that may be of little practical interest. This framing
 46 matters because the goal of a test campaign is frequently not to reconstruct
 47 the entire efficiency surface with uniform fidelity, but to determine the op-
 48 operationally decisive quantities—above all the location and value of the BEP
 49 and the shape of the high-efficiency region around it. A test plan optimized
 50 for uniform global accuracy and a test plan optimized for locating the BEP
 51 are not the same plan, and confusing the two can lead an otherwise careful
 52 study to measure the wrong points. Recognizing turbine performance test-
 53 ing as a targeted, budget-constrained decision process, rather than a passive
 54 data-collection exercise, is the conceptual starting point of this work.

55 *1.3. Research gap*

56 Data-driven and surrogate-based methods for reducing the number of
 57 evaluations in turbine engineering are now well established, but the existing
 58 literature addresses a different problem from the one posed here. The dom-
 59 inant body of work couples surrogate models—Gaussian process regression,
 60 polynomial response surfaces, or neural networks—to computational fluid
 61 dynamics (CFD) and uses them to accelerate the geometric design of run-
 62 ners and blades, validating the surrogate against further CFD rather than
 63 against physical measurements [4]. These studies typically operate over a

64 small number of geometric design variables and treat the expensive evaluation
 65 as a simulation, not a laboratory test. A second, more recent strand
 66 applies machine learning to experimental turbine data to predict efficiency,
 67 but it fits models to data that have already been collected and does not use
 68 the model to decide which experiments to run; the sampling plan is fixed in
 69 advance [5]. Classical design of experiments in turbine work—central com-
 70 posite designs, factorial designs, Latin hypercube designs—is likewise spec-
 71 ified before any testing begins and is non-adaptive [6]. What is absent is a
 72 study that treats the construction of a turbine performance map from real
 73 physical experiments as an adaptive, budget-constrained decision process,
 74 and that benchmarks an adaptive policy against the classical fixed designs
 75 on the specific task of locating the BEP. The present work addresses this gap
 76 directly. It uses a measured ground-truth dataset for a small-hydro propeller
 77 turbine, frames test-point selection as sequential decision-making, and com-
 78 pares adaptive Gaussian-process policies—Bayesian optimization and two
 79 active-learning designs—against random, grid, and Latin hypercube sam-
 80 pling across a sweep of experimental budgets, with both the full hill chart
 81 and the BEP as explicit decision targets.

82 *1.4. Contributions and organization*

83 The contributions of this paper are as follows.

- 84 1. We formulate the construction of a small-hydro propeller-turbine per-
 85 formance map as a sequential, budget-constrained decision problem
 86 with two distinct objectives—uniform surface reconstruction and best-
 87 efficiency-point identification—and we make explicit why these objec-
 88 tives demand different sampling strategies and different success metrics.
- 89 2. We benchmark six sampling policies—a Bayesian optimization (BO-
 90 UCB) policy, two Gaussian-process active-learning policies (ALM and
 91 ALC), and three classical design-of-experiments baselines (random, grid,
 92 and Latin hypercube sampling)—on a measured ground-truth dataset
 93 of 72 physical experiments, sweeping the budget from 6 to 30 experi-
 94 ments and evaluating every policy across 20 random seeds so that the
 95 comparison reflects expected behavior rather than a single fortunate
 96 run.
- 97 3. We show that the two objectives select opposite winners. The active
 98 learners reconstruct the full hill chart to $R^2 \geq 0.99$ from as few as 10
 99 experiments and to $R^2 \approx 0.996$ at 12 (an 83.3% reduction), while the

Bayesian policy is the weakest map-builder; conversely, the Bayesian policy locates the exact BEP node on every seed from 14 experiments (an 81% reduction), while every other policy plateaus near $\Delta\eta \approx 10^{-2}$. We confirm each separation with paired Wilcoxon signed-rank tests and argue that global accuracy is the wrong success criterion for a BEP-directed campaign.

4. We quantify the achievable experiment reduction through a budget sweep from 6 to 30 experiments, identify the very-low-budget regime ($n < 8$) in which model-based policies degrade and space-filling designs are the safer choice, and derive tiered practical guidance that matches the sampling policy to the deliverable.

The remainder of the paper is organized as follows. Section 2 reviews surrogate and machine-learning models for turbine performance, classical design of experiments, and Bayesian optimization as a framework for adaptive experimental design, and positions the present work against the closest prior art. Section 3 formulates the performance-mapping task as a sequential decision problem and defines the decision objective. Section 4 describes the experimental dataset, the Gaussian-process surrogate and the adaptive sampling policies, the initialization, the baseline designs, and the evaluation protocol. Section 5 reports the results. Section 6 discusses the findings, their decision-support implications, and the limitations of the study, and concludes.

2. Related Work

2.1. Surrogate and machine-learning models for turbine performance maps

Surrogate modeling has become a standard tool for reducing the computational cost of turbine analysis and design. The representative contribution in the hydropower context is the machine-learning surrogate of Masood et al. [4], who coupled a Gaussian process model to CFD to accelerate the simulation-driven optimization of a Kaplan turbine runner. Their surrogate is trained on CFD evaluations over a parametric geometric design space and is validated against further CFD; the reported outcome is a substantial improvement in the efficiency of the optimized design relative to the baseline geometry, achieved at a fraction of the computational cost of exhaustive simulation. This work is emblematic of a large family of studies in which the expensive evaluation is a numerical simulation, the design variables describe machine geometry, and the validation reference is itself computational. The

135 objective in these studies is design improvement, not the empirical character-
 136 ization of an existing machine from laboratory measurements. A more recent
 137 line of work brings machine learning to bear on experimental turbine data.
 138 Besni et al. [5] developed machine-learning models to predict the efficiency
 139 of Francis-type hydraulic turbines using data obtained from comprehensive
 140 model-acceptance performance testing, combining experimental and numeri-
 141 cal information in the predictive model. This is closer in spirit to the present
 142 study because the data are physical measurements rather than simulations.
 143 The decisive difference is that the predictive model is fitted to a sampling
 144 plan that has been fixed in advance; the model does not participate in de-
 145 ciding which operating points to test, and the number of experiments is not
 146 itself the object of optimization. The same observation applies to the broader
 147 literature on efficiency-map and hill-chart fitting, in which measured points
 148 are interpolated or regressed after the fact to yield a continuous character-
 149 istic. Beyond these representative cases, a rapidly growing body of work
 150 applies machine learning to the performance characterization of hydraulic
 151 machines. Data-driven models have been used to predict the best-efficiency
 152 operating point and the complete performance curves of pumps operating as
 153 turbines [7, 8], and to estimate turbine power output directly from operating
 154 data [9]. The broader role of artificial intelligence in turbomachinery de-
 155 sign, validation, and condition monitoring—including Gaussian-process sur-
 156 rogates and Bayesian optimization—has recently been surveyed [10], and
 157 hybrid schemes that combine neural networks with physics-based off-design
 158 analysis illustrate the same trend in adjacent turbomachinery [11]. In every
 159 case the surrogate is fitted to a predetermined dataset; it does not decide
 160 which experiments to perform.

161 *2.2. Classical design of experiments in turbine experimentation*

162 Design of experiments (DOE) provides the classical toolkit for choosing
 163 where to sample an input space when each observation is costly. In turbine
 164 and pump research the dominant approach is response surface methodol-
 165 ogy [12] built on central composite designs, which place experimental points
 166 at factorial corners, axial points, and a center point to support the fitting
 167 of a second-order polynomial model. Betancour et al. [6], for example, used
 168 a central composite design to optimize the runner of a gravitational vortex
 169 turbine, monitoring the response through CFD and validating the optimum
 170 experimentally; comparable central composite designs appear throughout the
 171 pump-as-turbine and mixed-flow literature. Latin hypercube sampling, in-

172 introduced by McKay et al. [13], is the other widely used design, valued for
 173 its space-filling properties and its even projection of points onto each input
 174 dimension. Full or fractional factorial grids and uniform random sampling
 175 complete the set of conventional choices. The defining feature of all of these
 176 designs is that they are non-adaptive: the complete set of sampling locations
 177 is determined before any measurement is taken, using only prior knowledge of
 178 the input ranges and no information from the responses themselves. This is
 179 appropriate when the goal is to characterize the whole input space uniformly
 180 or to fit a low-order global model, and it is the reason space-filling designs
 181 perform strongly on global reconstruction metrics. It is less appropriate when
 182 the goal is concentrated on a particular feature of the response, such as the
 183 location of an optimum, because the budget is spent uniformly rather than
 184 where the feature lies. Recent applications confirm the continued prevalence
 185 of these designs in energy and turbomachinery engineering: central com-
 186 posite designs coupled to computational fluid dynamics have been used to
 187 optimize turbine-duct geometry [14]. To soften the strictly non-adaptive na-
 188 ture of classical space-filling designs, adaptive Latin hypercube schemes have
 189 been proposed that add samples sequentially in regions of high surrogate de-
 190 viation [15]; such schemes move partway toward the adaptive policy studied
 191 here while remaining variance-driven rather than objective-driven.

192 *2.3. Bayesian optimization and adaptive experimental design*

193 Bayesian optimization (BO) is a framework for the sequential, model-
 194 based optimization of expensive black-box functions [16, 17]. It maintains a
 195 probabilistic surrogate of the unknown function—most commonly a Gaussian
 196 process [18]—and uses an acquisition function derived from the surrogate’s
 197 posterior to decide where to evaluate next, balancing exploitation of regions
 198 believed to be near-optimal against exploration of regions where the surro-
 199 gate is uncertain. The Expected Improvement criterion of Jones et al. [16]
 200 and the upper-confidence-bound family [19] are the canonical acquisition
 201 functions, and the approach has been applied across machine-learning hy-
 202 perparameter tuning [20] and a wide range of engineering design tasks. Of
 203 particular relevance here is the use of BO not for optimization per se but as
 204 a policy for adaptive experimental design. Greenhill et al. [21] review this
 205 perspective explicitly, framing BO as an automated decision policy for choos-
 206 ing experiments and contrasting it with classical fixed designs; benchmark
 207 studies in experimental materials science [22] likewise compare BO against
 208 space-filling and random designs across diverse real datasets. The consis-

209 tent finding across this literature is that adaptive policies excel when the
 210 objective is to find an optimum efficiently, precisely because they deliberately
 211 concentrate sampling near the optimum and decline to spend budget
 212 on uninformative regions—at the expense of global model accuracy. This exploitation
 213 behavior is central to interpreting the results of the present study.
 214 Within fluid machinery specifically, Bayesian optimization and Gaussian-
 215 process surrogates have been applied to one-dimensional turbomachinery design
 216 through transfer learning [23], to multi-objective aerodynamic shape
 217 optimization of airfoils [24], to coupled aerodynamic and aeroacoustic blade
 218 design [25], and to high-dimensional aero-structural compressor-blade design [26].
 219 More generally, active-learning strategies that construct surrogate models with the
 220 fewest possible expensive evaluations [27] share the objective-directed sampling
 221 philosophy adopted here. These applications are predominantly simulation-based and
 222 geometry-oriented; their adaptation to the selection of physical operating points on
 223 a measured turbine, with the BEP as the explicit target, is the contribution of the
 224 present work.

225 *2.4. Positioning of the present work*

226 Table 1 positions the present study against the closest prior art along
 227 four axes: the nature of the data, the method, the optimization target, and
 228 whether the approach is benchmarked against classical designs. The combination
 229 addressed here—physical experimental data, adaptive Gaussian-process policies
 230 (Bayesian optimization and active learning) benchmarked against random, grid,
 231 and Latin hypercube designs, on a small-hydro propeller turbine, with the BEP
 232 as the explicit target—does not appear in the existing literature. The present
 233 work does not claim to be the first application of Bayesian optimization to a
 234 turbine; it claims novelty in this specific combination and, more importantly,
 235 in the decision-level finding that the sampling policy must be matched to the
 236 deliverable: exploitation-driven sampling wins for BEP identification despite
 237 inferior global metrics, while uncertainty-driven active learning wins for hill-
 238 chart reconstruction.

239 **3. Problem Formulation**

240 *3.1. Performance map and best efficiency point*

241 Let the operating condition of the turbine be described by the discharge
 242 Q and the head H , and let $\eta(Q, H)$ denote the measured efficiency at that
 243 condition. The performance map is the function η defined over the feasible

Table 1: Positioning relative to the closest prior work.

Study	Data source	Method	Target	Benchmarked vs. classical DOE
Masood et al. (2021)	CFD	GP surrogate + optimization	Runner geometry design	No
Besni et al. (2025)	Experiment (model test)	ML efficiency prediction	Efficiency prediction	No
Betancour et al. (2023)	CFD + experiment	RSM / CCD	Runner geometry design	Single design family
Greenhill et al. (2020)	Various (review)	BO / adaptive design	Generic optimization	Conceptual
This work	Experiment (test rig)	Adaptive GP policies (BO, ALM, ALC)	BEP localization + map	Random, Grid, LHS

operating domain $\mathcal{X}$, a subset of the (Q, H) plane, which in the present study spans $Q \in [0.791, 1.123] \text{ m}^3 \text{ s}^{-1}$ and $H \in [15, 22] \text{ m}$. The best efficiency point is the maximizer of efficiency over this domain:

$$(Q^*, H^*) = \arg \max_{(Q, H) \in \mathcal{X}} \eta(Q, H), \quad \eta^* = \eta(Q^*, H^*). \quad (1)$$

In the measured ground-truth dataset the BEP is located at $Q = 0.8185 \text{ m}^3 \text{ s}^{-1}$, $H = 16.00 \text{ m}$ (guide-vane angle 37.5°), with $\eta^* = 0.6973$, in the low-discharge, low-head corner of the envelope where the efficiency ridge is highest. The efficiency surface is smooth and unimodal in the high-efficiency region, with a single broad ridge descending toward a low-efficiency corner at high discharge (efficiency falling to 0.497); this structure is typical of low-head propeller machines. Fig. 1 presents the measured efficiency surface of this turbine in three dimensions, and Fig. 2 shows the same data as a contour map of the discharge-head plane with the BEP marked.

The true function η is observable only through physical experiments. Each experiment at a chosen condition (Q_i, H_i) returns a measurement $y_i = \eta(Q_i, H_i) + e_i$, where e_i captures measurement and repeatability error. Experiments are expensive, so the number that can be performed is limited to a budget N that is small relative to the resolution at which the map would ideally be known.

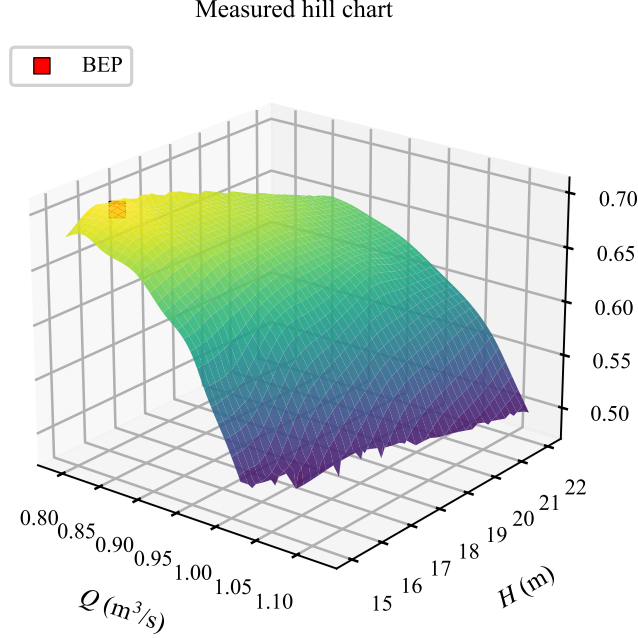


Figure 1: Measured ground-truth efficiency surface of the 137 kW propeller turbine (72 experimental points) in the discharge-head plane.

3.2. Sequential decision formulation

We cast the construction of the performance map as a sequential decision problem. After t experiments the state is the set of observed condition–efficiency pairs $\mathcal{D}_t = \{(Q_i, H_i, y_i)\}_{i=1}^t$. An action is the choice of the next operating condition $(Q_{t+1}, H_{t+1}) \in \mathcal{X}$ at which to run an experiment; each operating point is specified by (Q, H) , which fixes the corresponding guide-vane setting on the test manifold. A policy π is a rule that maps the current state to the next action, $(Q_{t+1}, H_{t+1}) = \pi(\mathcal{D}_t)$. The process begins from an initial design of n_0 points and proceeds until the budget N is exhausted. This formulation unifies the methods compared in this paper. The classical designs correspond to non-adaptive policies whose actions do not depend on the observed efficiencies: random sampling draws each condition independently from $\mathcal{X}$, grid sampling places conditions on a fixed lattice, and Latin hypercube sampling selects a space-filling set with even one-dimensional projections. Bayesian optimization corresponds to an adaptive policy: it maintains a surrogate model of η given $\mathcal{D}_t$ and chooses each action by maximizing

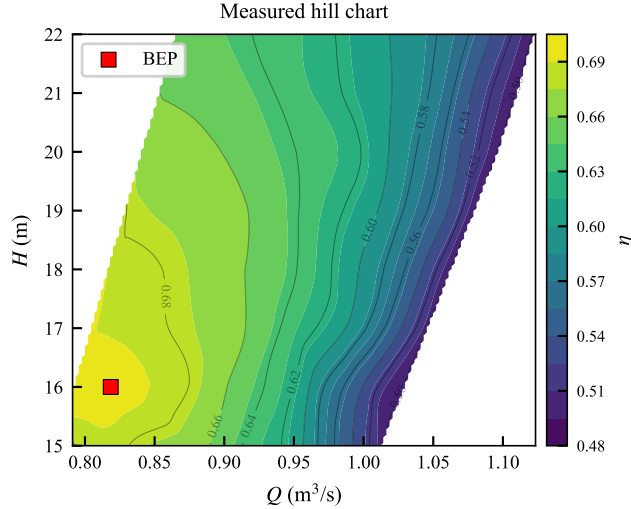


Figure 2: Measured ground-truth efficiency contour map with the experimental points and the best efficiency point (BEP).

an acquisition function that depends on the surrogate’s posterior mean and variance, so that the next experiment is selected in light of everything observed so far. The distinction between non-adaptive and adaptive policies is the central methodological variable of the study.

3.3. Decision objective and the inadequacy of global fit

The choice of objective determines which policy is good. Two objectives must be distinguished. The first is global reconstruction: to recover the entire surface η over $\mathcal{X}$ with uniform accuracy, measured by aggregate error statistics such as the mean absolute error or the coefficient of determination computed over the whole domain. The second is BEP identification: to determine (Q^*, H^*) and η^* , and to characterize the high-efficiency region around the optimum, as accurately as possible within the budget. These objectives are not aligned. A policy optimized for global reconstruction spreads its budget evenly and is rewarded for accuracy in regions of the operating envelope—such as the low-efficiency, high-discharge corner—that are of little operational interest. A policy optimized for BEP identification concentrates its budget near the optimum and is content to model the uninteresting regions poorly. Consequently, the global error of a BEP-directed policy can be large even when its localization of the optimum is exact, and conversely

297 a space-filling design can achieve excellent global error while placing no ex-
 298 periments close enough to the true optimum to resolve it precisely. The
 299 results of Section 5 demonstrate exactly this reversal. It follows that global
 300 reconstruction metrics are the wrong success criterion for a BEP-directed
 301 campaign. We therefore evaluate every policy under both objectives—global
 302 reconstruction metrics for the hill chart and localization error for the BEP—
 303 and rank each policy only under the objective it is designed to serve. The
 304 formal definitions of these metrics are given in Section 4.5.

305 4. Methodology

306 4.1. *Experimental dataset and test rig*

307 The ground-truth dataset consists of 72 measured operating points for a
 308 small-hydro propeller turbine, organized as eight constant-head test series at
 309 heads of 15, 16, 17, 18, 19, 20, 21, and 22 m, each comprising a sweep of guide-
 310 vane openings and the corresponding discharge. The measured variables at
 311 each point are the guide-vane angle, the discharge, the generator power,
 312 the mechanical power, and the resulting overall efficiency. Discharge spans
 313 0.791 to $1.123 \text{ m}^3 \text{ s}^{-1}$ and overall efficiency spans 0.497 to 0.6973 across the
 314 envelope. These 72 points constitute the reference against which all sampling
 315 policies are evaluated and are treated throughout as ground truth.

316 The turbine under study is an axial-flow propeller (bulb) turbine devel-
 317 oped for low-head small-hydro applications in Thailand under the Electricity
 318 Generating Authority of Thailand (EGAT) Small Hydro Power Research and
 319 Development Project, the same machine family later summarized by Phitak-
 320 surachai et al. [2]. The specific unit analyzed here is the 137 kW-rated turbine
 321 designed for the 15–20 m head class. The runner has four fixed blades of 0.4 m
 322 diameter with a hub-to-tip ratio of 0.35, fixed to the hub at 40° in the radial
 323 plane; the blade geometry is a marine-propeller form with a reversed mean
 324 camber line derived from a NACA 4410 airfoil section. Flow is regulated by
 325 twelve adjustable guide vanes, and the machine is a reaction turbine fitted
 326 with a draft tube. The turbine shaft is directly coupled, within the bulb, to a
 327 160 kW induction generator running at 1000 rpm. The sub-megawatt rating
 328 and low-head design place the unit firmly in the small-hydro class. Fig. 3
 329 shows the unit installed on the performance test rig.

330 Performance testing was carried out by EGAT’s Power Plant Performance
 331 Test Department (Production Efficiency Division) on a purpose-built rig at
 332 the Kaeng Krachan Dam site, Phetchaburi Province. At each constant head



Figure 3: The 137 kW axial-flow bulb turbine installed on the EGAT performance test rig at the Kaeng Krachan Dam site. Flow enters from the penstock at right through the bolted-flange casing; the bulb houses the directly coupled induction generator, and the draft-tube cone extends to the left. At each operating point the discharge, the generator and mechanical power, and the guide-vane angle were recorded through the oscilloscope and data-acquisition instrumentation visible in the foreground. Bringing the rig to a stable condition and averaging repeated readings at every operating point is the labor-intensive procedure that makes each of the 72 measured points costly and motivates reducing their number.

333 the guide-vane angle was swept while the discharge and the generator and
 334 mechanical power were recorded; the overall (system) efficiency reported in
 335 the dataset is the wire-to-water efficiency, equal to the turbine mechanical
 336 efficiency multiplied by an assumed induction-generator efficiency of approx-
 337 imately 0.90. The dataset also records the turbine mechanical efficiency sep-
 338 arately. At the BEP both efficiency definitions identify the same operating
 339 point, so the localization analysis that follows is unaffected by the choice be-
 340 tween them. The efficiency magnitudes in this dataset are consistent with the
 341 source documentation once two definitional points are recognized. First, the
 342 values analyzed here are overall system (wire-to-water) efficiencies, whereas
 343 the turbine mechanical efficiencies reported for the same machine are higher:

344 at the BEP the dataset records an overall efficiency of 0.697 and a mechanical
 345 efficiency of 0.777, the ratio being the assumed generator efficiency of about
 346 0.90. The 78–80% figures quoted in the turbine-design literature for this
 347 machine family refer to mechanical (turbine) efficiency, which the present
 348 dataset reproduces at 0.777. Second, the source performance tests computed
 349 efficiency from inlet pressure measured at two alternative planes—before the
 350 bulb, and after the bulb just upstream of the guide vanes—giving maximum
 351 turbine efficiencies of 75.17% (at 17 m head) and 77.93% (at 16 m head,
 352 guide-vane angle 37.5° , discharge $0.82 \text{ m}^3 \text{ s}^{-1}$) respectively for this 137 kW
 353 unit. These are the same measured runs evaluated across different measure-
 354 ment planes, not different test campaigns, and the 16 m, 37.5° , $0.82 \text{ m}^3 \text{ s}^{-1}$
 355 peak coincides with the BEP identified in the present dataset. The dataset
 356 analyzed here is therefore the EGAT 137 kW small-hydro propeller turbine
 357 campaign, with efficiency computed on the after-bulb plane, the configura-
 358 tion that yields the higher efficiency and whose mechanical-efficiency peak
 359 of 0.777 is consistent with the 77.93% reported for that plane. The BEP of
 360 the ground-truth dataset lies at $Q = 0.8185 \text{ m}^3 \text{ s}^{-1}$, $H = 16.00 \text{ m}$ (guide-vane
 361 angle 37.5°), $\eta^* = 0.6973$, which is the point of maximum measured overall
 362 efficiency. We note here, and return to it in Section 6.3, that the BEP head
 363 coincides exactly with an integer head node of the test matrix; this coinci-
 364 dence has consequences for the comparison that must be acknowledged.

365 4.2. Gaussian-process surrogate and adaptive sampling policies

366 The adaptive policy is a Bayesian optimization loop built on a Gaussian
 367 process (GP) surrogate of the efficiency function [18], implemented with the
 368 scikit-learn library [28]. The GP places a prior over functions specified by a
 369 constant mean and a product covariance kernel,

$$k(\mathbf{x}, \mathbf{x}') = C \cdot k_{\text{Matérn}}(\mathbf{x}, \mathbf{x}'; \nu = 5/2), \quad (2)$$

370 where C is a constant (signal-variance) kernel and $k_{\text{Matérn}}$ is the Matérn ker-
 371 nel with smoothness parameter $\nu = 5/2$ and a single shared length scale. The
 372 Matérn-5/2 kernel is preferred over the squared-exponential kernel because it
 373 imposes weaker smoothness assumptions appropriate to measured data [29].
 374 The kernel hyperparameters are obtained by maximizing the marginal likeli-
 375 hood with five random restarts of the optimizer; a small jitter ($\alpha = 1 \times 10^{-6}$)
 376 is added to the diagonal for numerical stability, and the efficiency response
 377 is standardized internally. The surrogate takes three standardized inputs—
 378 discharge Q , head H , and guide-vane angle—and all six sampling policies use

the same three inputs and GP configuration, so the comparison isolates the sampling strategy alone. The guide-vane angle is the actuated control variable in this campaign: each (Q, H) condition maps to a unique guide-vane setting—the within-head Pearson correlation between guide-vane angle and discharge, averaged over the eight test heads, is 0.986 (range 0.983–0.988)—so an operating point is fully specified by (Q, H) , and the sampling policies operate in the (Q, H) plane. Reconstruction accuracy is evaluated at all 72 measured operating points, and the efficiency field is visualized on the conventional (Q, H) hill-chart plane.

Trained on the sampled set S with inputs X_S and measured efficiencies y_S , the Gaussian process yields a posterior at any operating point $\mathbf{x}$ that is itself Gaussian, with closed-form mean and variance

$$\mu_S(\mathbf{x}) = m + k(\mathbf{x}, X_S) [K_S + \sigma_n^2 I]^{-1} (y_S - m), \quad (3)$$

$$\sigma_S^2(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - k(\mathbf{x}, X_S) [K_S + \sigma_n^2 I]^{-1} k(X_S, \mathbf{x}), \quad (4)$$

where $K_S = k(X_S, X_S)$ and σ_n^2 is the small jitter term. The posterior mean μ_S is the reconstructed efficiency surface, while the posterior variance σ_S^2 is the surrogate’s own admission of how uncertain it remains. Crucially, the variance in Eq. (4) depends only on the *locations* of the observations and not on their measured values, so the informativeness of a candidate operating point can be scored in closed form before the experiment is run—the property the active-learning acquisitions exploit.

The three adaptive policies (BO-UCB, ALM, and ALC) share a single greedy loop and differ only in the acquisition function $a(\mathbf{x})$: (i) initialize S with n_0 points drawn at random; (ii) fit the GP on (X_S, y_S) and evaluate μ_S and σ_S from Eqs. (3)–(4); (iii) select $\mathbf{x}_{\text{next}} = \arg \max_{\mathbf{x} \in \mathcal{X} \setminus S} a(\mathbf{x})$; (iv) run the experiment, append it to S , and repeat (ii)–(iv) until $|S| = n$. The three acquisitions encode different objectives, developed next.

Given the posterior mean $\mu_t(\mathbf{x})$ and posterior standard deviation $s_t(\mathbf{x})$ after t observations, BO-UCB chooses the next experiment by maximizing an upper confidence bound acquisition function over the candidate operating points,

$$a(\mathbf{x}) = \mu_t(\mathbf{x}) + \kappa s_t(\mathbf{x}), \quad \kappa = 2.0. \quad (5)$$

The upper-confidence-bound criterion is used because it provides an explicit and interpretable trade-off between exploiting points of high predicted efficiency (the μ term) and exploring points of high posterior uncertainty (the s term), with $\kappa = 2.0$ weighting exploration moderately. At each iteration

the GP is refitted on all points observed so far, the acquisition function is evaluated over all not-yet-sampled operating points, and the point of maximum acquisition is selected for the next experiment, which is realized by reading the corresponding ground-truth efficiency. The loop runs, refitting the surrogate after every observation, until the experimental budget n is exhausted.

Two Gaussian-process active-learning policies complete the set of adaptive methods. They share the surrogate, the initialization, and the sequential loop of the Bayesian policy, and differ only in the acquisition function: whereas the UCB acquisition is built to find a maximum, the active-learning acquisitions are built to make the surrogate accurate everywhere, and the contrast between these two design intents is the methodological axis of the study. Active Learning MacKay (ALM) [30] selects the candidate operating point at which the posterior standard deviation is largest,

$$a(\mathbf{x}) = s_t(\mathbf{x}), \quad (6)$$

so that each experiment removes the largest remaining pocket of predictive uncertainty. Under a Gaussian process this greedy rule maximizes the information gained about the function value at the queried point, and it tends to produce a near space-filling pattern with a mild bias toward the domain boundary, where the posterior variance is naturally highest. Active Learning Cohn (ALC) [31] instead selects the candidate whose observation most reduces the surrogate’s integrated posterior variance over the whole candidate set,

$$a(\mathbf{x}) = -\frac{1}{N} \sum_{i=1}^N s_{t+\mathbf{x}}^2(\mathbf{x}_i), \quad (7)$$

where the augmented posterior variance is evaluated at each of the N candidate points. Because the Gaussian-process posterior variance depends only on the locations of the observations and not on their measured values, the effect of adding a candidate can be scored in closed form before the experiment is run, with the fitted kernel held fixed, at a per-step cost of $O(N)$ candidate evaluations for ALM and UCB and $O(N^2)$ for ALC. ALC is the sequential, greedy counterpart of the integrated-mean-square-error optimal designs of classical computer-experiment theory [32].

4.3. Initialization

Every sequential policy (BO-UCB, ALM, and ALC) is initialized with $n_0 = 5$ operating points drawn uniformly at random, without replacement,

445 from the measured candidate set, after which the acquisition function gov-
 446 erns all subsequent selections. A random initial design makes the sequential
 447 policies stochastic, so their run-to-run variability is characterized across ran-
 448 dom seeds on the same footing as the random and Latin hypercube baselines
 449 (Section 4.5). Structured alternatives, such as the corners-and-center core of
 450 a classical central composite design, are a natural refinement; a systematic
 451 comparison of initialization schemes is identified as future work in Section 6.4.

452 4.4. Baseline design-of-experiments methods

453 Three classical, non-adaptive designs serve as baselines, each allocated
 454 the same total experimental budget n as the adaptive policies to ensure a
 455 fair comparison.

456 *Random sampling* draws n operating conditions without replacement from
 457 the measured points. It provides an unbiased but uncontrolled coverage of
 458 the space and serves as the minimal baseline.

459 *Grid sampling* constructs a regular lattice spanning the discharge and
 460 head ranges and snaps each lattice node to the nearest measured operating
 461 point, reproducing the conventional practice of testing at evenly spaced op-
 462 erating conditions; when the budget is not a perfect square, the selection is
 463 completed to n with randomly chosen unused points, so grid sampling is also
 464 mildly stochastic at such budgets.

465 *Latin hypercube sampling* [13] generates a two-dimensional space-filling
 466 design over the discharge–head ranges with even one-dimensional stratifica-
 467 tion, snaps each sample to the nearest measured point, and fills any shortfall
 468 with randomly chosen unused points.

469 For every design, the sampled conditions are evaluated against the ground-
 470 truth dataset, a surrogate is fitted to the resulting observations using the
 471 same Gaussian process specification as in the preceding subsection to en-
 472 sure that differences in performance arise from the sampling policy and not
 473 from the interpolating model, and the fitted surrogate is used both to recon-
 474 struct the surface and to estimate the BEP. All six policies retain a stochas-
 475 tic element—the sequential policies through their random initialization, grid
 476 sampling through its random completion at non-square budgets, and ran-
 477 dom and Latin hypercube sampling by construction—and all are therefore
 478 evaluated across random seeds (Section 4.5).

479 4.5. Evaluation protocol

480 All policies are evaluated at a series of common budgets n swept from
 481 6 to 30 experiments against the complete measured dataset of 72 points;
 482 a budget of n corresponds to a reduction of $(72 - n)/72$ in the number of
 483 physical experiments, from 58.3% at $n = 30$ to 91.7% at $n = 6$. The budget
 484 $n = 12$ (an 83.3% reduction) serves as the headline operating point for the
 485 tables and illustrative figures, chosen by the engineering criterion stated in
 486 Section 5.3.

487 *Global reconstruction metrics.* The reconstructed surface is compared
 488 to the ground truth over all 72 experimental points using the mean abso-
 489 lute error (MAE), the root-mean-square error (RMSE), the coefficient of de-
 490 termination (R^2), and the mean absolute percentage error (MAPE). These
 491 metrics rank the policies under the global-reconstruction objective; under
 492 the localization objective they are reported only to characterize each policy’s
 493 behavior.

494 *BEP localization metrics.* The primary metrics are the absolute errors in
 495 the predicted BEP location and value relative to ground truth: the discharge
 496 error $|\hat{Q} - Q^*|$, the head error $|\hat{H} - H^*|$, and the efficiency error $|\hat{\eta} - \eta^*|$. To
 497 avoid crediting a policy merely for sampling the peak, the BEP is read from
 498 the GP prediction over the full 72-point grid, not from the sampled rows.

499 *Statistical protocol.* Because every policy retains a stochastic element, all
 500 metrics are computed across 20 independent random seeds and reported as
 501 mean $\pm$ standard error; each seed fixes the random initialization of the se-
 502 quential policies and the draws of the random and Latin hypercube designs.
 503 Pairwise differences between policies are tested with one-sided Wilcoxon
 504 signed-rank tests, pairing the methods on the seeds as matched replications;
 505 with 20 seeds the smallest attainable exact p -value is $2^{-20} \approx 10^{-6}$.

506 *Budget analysis.* To quantify the achievable experiment reduction, both
 507 the global reconstruction error and the BEP localization error are reported
 508 as functions of the budget n over the range 6 to 30 experiments, and the
 509 smallest budget at which the Bayesian policy locates the exact BEP node on
 510 every seed is determined.

511 5. Results

512 The results are organized around the two objectives defined in Section 3.
 513 Section 5.1 reports global hill-chart reconstruction; Section 5.2 reports best-
 514 efficiency-point localization; Section 5.3 presents the budget sweep that is the

core evidence of the study; Section 5.4 gives the significance tests; Section 5.5 examines the very-low-budget regime; Section 5.6 verifies that the surrogate’s posterior uncertainty tracks its actual error; and Section 5.7 distills the findings into practical guidance for planning a reduced test campaign. All policies are averaged over 20 seeds and reported as mean $\pm$ standard error. Unless stated otherwise the illustrative budget is $n = 12$, an 83.3% reduction from the 72-point campaign (justified in Section 5.3).

5.1. Global hill-chart reconstruction

For recovering the entire efficiency map, the model-based active learners are the strongest methods and the optimizer is the weakest. Table 2 reports reconstruction and localization metrics for all six policies at $n = 12$. The integrated-variance active learner ALC attains the best global accuracy ($R^2 = 0.9965$, MAE = 0.00218, MAPE = 0.35%), with the maximum-variance learner ALM statistically indistinguishable from it ($R^2 = 0.9961$; the difference is not significant, Section 5.4). The space-filling designs follow (LHS $R^2 = 0.9945$, grid $R^2 = 0.9938$), while random sampling ($R^2 = 0.9380$) and, notably, Bayesian optimization ($R^2 = 0.9207$) are markedly worse. BO-UCB is the worst map-builder of the six: by design it concentrates its budget near the predicted optimum and starves the rest of the domain, so its surrogate is accurate near the peak but biased elsewhere—the expected behavior of an optimization acquisition applied to a reconstruction task. Fig. 4 shows the reconstructed hill chart obtained by each of the six policies at $n = 12$, and Fig. 5 presents the same reconstructions as three-dimensional efficiency surfaces at a shared viewing angle. Fig. 6 maps the absolute reconstruction error for each policy over the operating plane, and Fig. 7 presents the same error as three-dimensional surfaces, on which the tall error ridge over the low-efficiency, high-discharge corner produced by random sampling and BO-UCB is immediately visible. Fig. 8 then contrasts where each policy places its 12 experiments. Across these views the active learners and space-filling designs cover the whole envelope, whereas BO-UCB leaves the low-efficiency, high-discharge corner sparse and reconstructs it poorly.

5.2. Best-efficiency-point localization

For the opposite objective—pinpointing the BEP—the ranking inverts. The rightmost columns of Table 2 report the localization errors, and here Bayesian optimization is decisively best: at $n = 12$ it already matches the ground-truth head exactly ($\Delta H = 0$) and reaches an efficiency error

Table 2: Reconstruction and BEP-localization metrics at $n = 12$ (83.3% reduction), mean over 20 seeds. Best value in each column in bold. Reconstruction: lower MAE/RMSE/MAPE and higher R^2 are better. Localization: lower $\Delta Q/\Delta H/\Delta \eta$ are better.

Method	MAE	RMSE	R^2	MAPE (%)	ΔQ	ΔH	$\Delta \eta$
ALC	0.00218	0.00354	0.9965	0.35	0.0235	0.50	0.00714
ALM	0.00239	0.00380	0.9961	0.38	0.0131	0.75	0.01015
LHS	0.00272	0.00445	0.9945	0.44	0.0215	0.80	0.00748
Grid	0.00313	0.00476	0.9938	0.50	0.0108	0.70	0.01131
Random	0.00609	0.00988	0.9380	1.07	0.0230	0.55	0.00861
BO-UCB	0.00986	0.01577	0.9207	1.77	0.0137	0.00	0.00030

551 $\Delta \eta = 0.0003$, one to two orders of magnitude below every other method ($\Delta \eta$
 552 between 0.007 and 0.011 for the active learners and space-filling designs). By
 553 $n = 14$ the UCB policy locates the exact BEP node, with discharge, head,
 554 and efficiency error all at the level of numerical noise ($\Delta \eta \rightarrow 0$). This fol-
 555 lows directly from the exploitation term of the UCB acquisition: it climbs
 556 the efficiency ridge and expends its budget resolving the peak—exactly what
 557 BEP localization rewards and exactly what penalized it on the global metric.
 558 The active learners and space-filling designs, spreading their budget for cov-
 559 erage, never resolve the optimum as sharply and plateau near 10^{-2} . Fig. 9
 560 juxtaposes the two behaviors at the same budget in a single frame.

561 5.3. Experiment budget versus accuracy

562 The central evidence of the study is the budget sweep in Fig. 10, which
 563 reports both objectives as a function of the number of experiments n from 6 to
 564 30. For reconstruction (panel a) the active learners reach engineering-grade
 565 accuracy very early: ALC attains $R^2 = 0.994$ by $n = 10$ and $R^2 = 0.996$ by
 566 $n = 12$, and ALM tracks it closely, while BO-UCB never exceeds $R^2 \approx 0.96$
 567 anywhere in the range. For localization (panel b) BO-UCB drives the BEP
 568 efficiency error to 0.0003 by $n = 12$ and to the level of numerical noise
 569 (below 4×10^{-6}) by $n = 14$, whereas the other methods remain near 10^{-2}
 570 throughout. At large budgets ($n \geq 25$) all reconstruction methods saturate
 571 above $R^2 = 0.99$ and the differences vanish; the value of the comparison is
 572 visible only in the low-budget regime, which is precisely where experiments
 573 are scarce.

574 This motivates $n = 12$ as the headline budget for the illustrative figures,

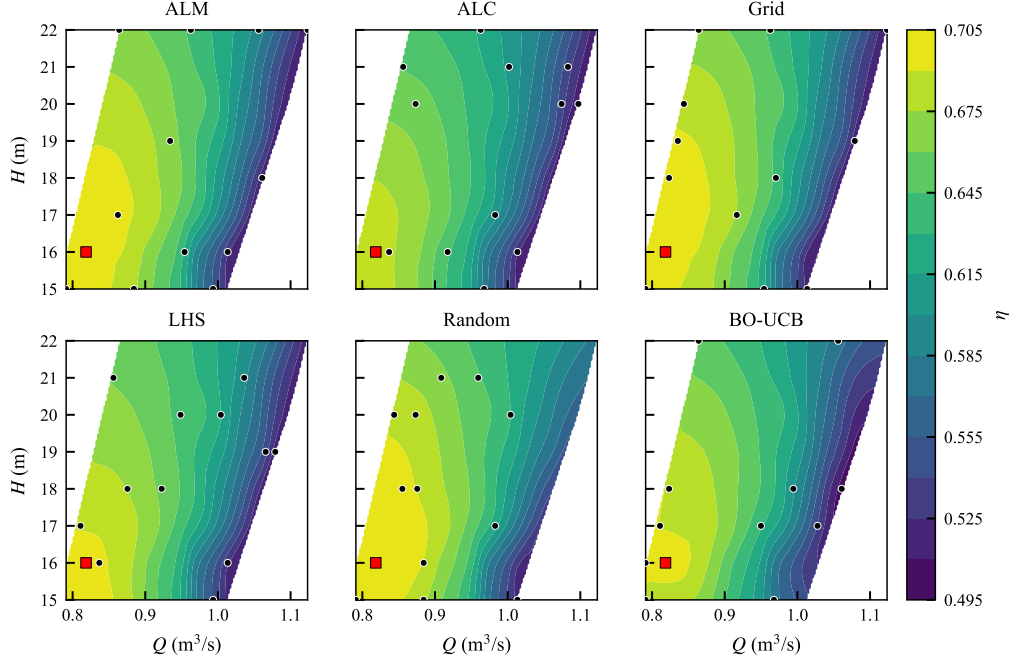


Figure 4: Reconstructed hill-chart contours at $n = 12$ for all six policies (the 12 sampled points in black, the true BEP marked by a red square). The active learners and space-filling designs recover the whole surface, whereas BO-UCB distorts the low-efficiency, high-discharge region.

575 chosen by an explicit engineering criterion—the smallest budget at which
 576 the recommended reconstruction method reaches $R^2 \geq 0.99$ and $\text{MAPE} <$
 577 0.5% —rather than by inspection. ALC clears this bar at $n = 10$ ($R^2 = 0.994$,
 578 $\text{MAPE} = 0.46\%$) and ALM at $n = 12$ ($R^2 = 0.996$, $\text{MAPE} = 0.38\%$); we
 579 adopt $n = 12$, the conservative point at which both recommended learners
 580 clear the bar, an 83.3% reduction from the full campaign.

581 The headline reduction result follows directly. For the full hill chart,
 582 active learning attains $R^2 \approx 0.996$ from just 12 of the 72 measured points—
 583 an 83% reduction—and reaches $R^2 \geq 0.99$ from as few as 10 points. For the
 584 BEP alone, Bayesian optimization identifies the exact optimum node using
 585 14 experiments, an 81% reduction. In both cases the deliverable is obtained
 586 at well under a quarter of the original test effort, provided the sampling
 587 policy is matched to the objective.

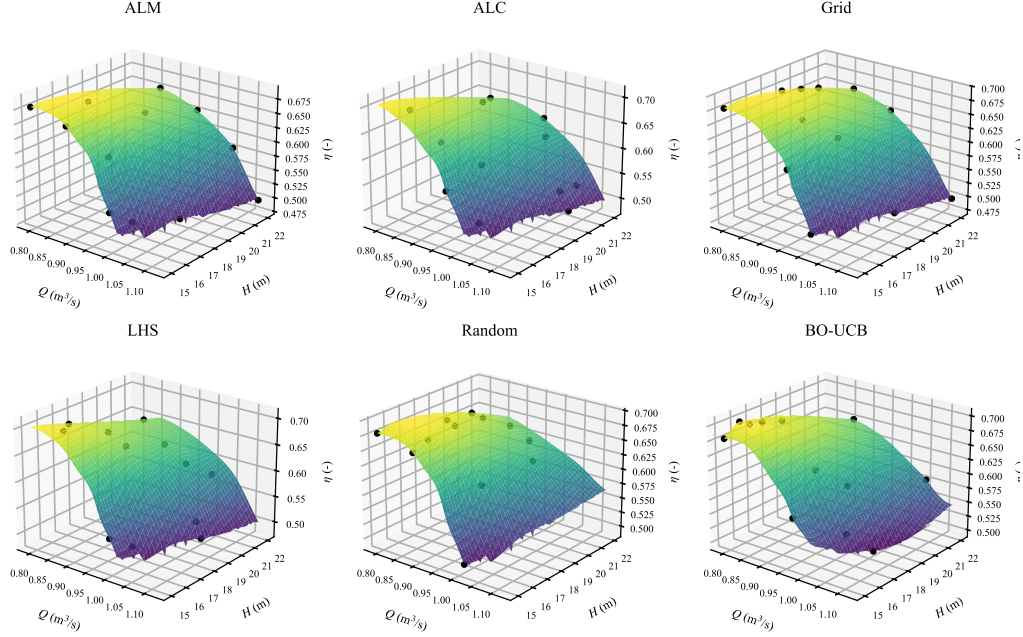


Figure 5: Reconstructed efficiency surfaces at $n = 12$ for all six policies, rendered at the shared viewing angle used throughout for comparability. Sampled points are shown in black.

5.4. Statistical significance

Because these averages could reflect seed-to-seed luck, each headline claim is tested with a Wilcoxon signed-rank test, pairing methods on the 20 seeds as matched replications (Table 3). For reconstruction (reference ALC at $n = 12$) ALC significantly outperforms BO-UCB ($p \approx 10^{-6}$), random ($p \approx 10^{-5}$), grid ($p = 1.3 \times 10^{-4}$) and LHS ($p = 0.009$), while the ALC–ALM difference is not significant ($p = 0.09$): the two active learners are statistically equivalent. For localization (reference BO-UCB at $n = 12$) BO-UCB beats every other method on efficiency error at $p \leq 4.8 \times 10^{-6}$, at or near the smallest p attainable with 20 seeds, a floor small enough that its localization advantage is unambiguous. Read across budgets, ALC and ALM significantly beat BO-UCB and random at essentially every $n \geq 8$ and beat grid and LHS from $n = 12$ upward; the ALM–ALC difference is never significant, so we recommend ALM as the practical default for reconstruction—equal accuracy at lower computational cost. Fig. 11 visualizes the ranking inversion between the two objectives at $n = 12$.

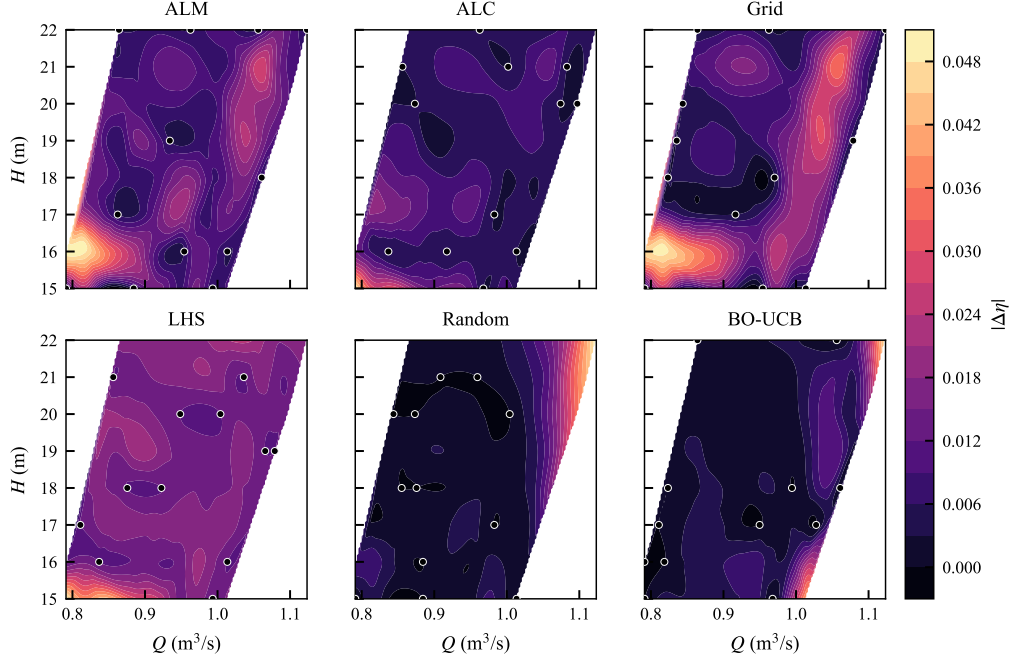


Figure 6: Absolute reconstruction error $|\Delta\eta|$ at $n = 12$ for all six policies. The error concentrates where a policy leaves the operating envelope sparsely sampled.

5.5. The very-low-budget regime

One qualification bounds the recommendation. Below about eight experiments the Gaussian process is fitted on too few points for the model-based acquisitions to be reliable, and the integrated-variance learner in particular can degrade: at $n = 6$, ALC drops to $R^2 = 0.82$ and BO-UCB to $R^2 = 0.84$, whereas the space-filling and maximum-variance designs remain robust (LHS $R^2 = 0.97$, grid $R^2 = 0.96$, ALM $R^2 = 0.97$). In this extreme-reduction corner a model-free space-filling design is the safer choice. The practical guidance is therefore tiered: use Latin hypercube sampling for the smallest budgets ($n < 8$) and switch to active learning (ALM or ALC) once $n \geq 10$, where it reliably dominates. For BEP localization the recommendation is unqualified across the range examined: Bayesian optimization is preferred at every budget.

In summary, the six policies separate cleanly into three groups matched to purpose: the active learners (ALC, ALM) for reconstructing the whole hill chart, Bayesian optimization for locating the BEP, and space-filling designs

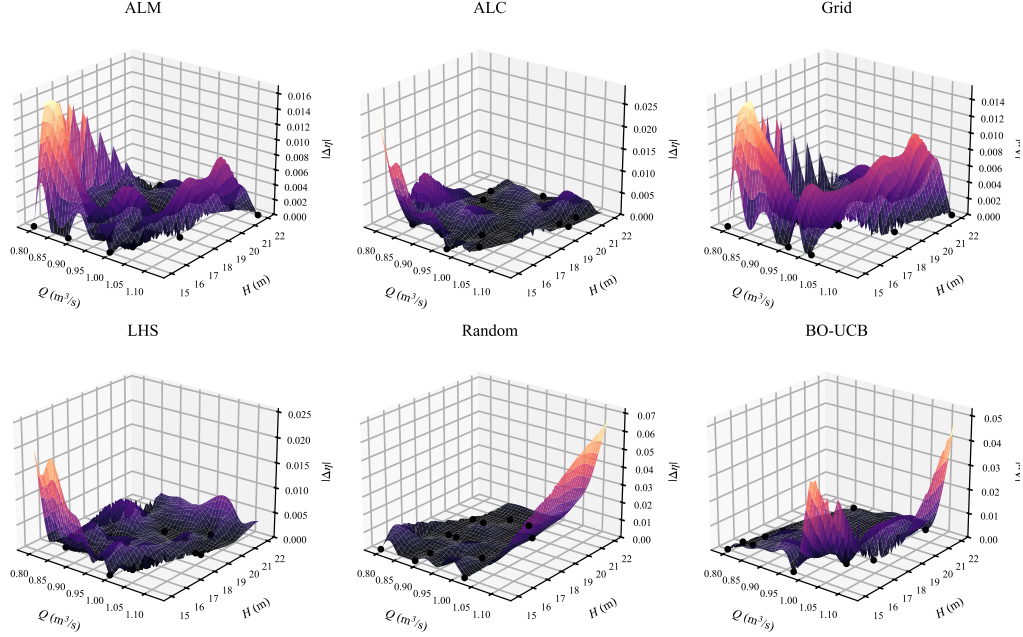


Figure 7: Absolute reconstruction error $|\Delta\eta|$ at $n = 12$ for all six policies as three-dimensional surfaces (the companion to Fig. 6). The vertical scale exposes how much larger the peak error is for random sampling and BO-UCB—a tall ridge over the low-efficiency, high-discharge corner—than for the active learners and space-filling designs.

620 (chiefly LHS) as the robust fallback at the very lowest budgets. Both recom-
 621 mended tools are data-driven and share the same surrogate, so the practical
 622 cost of choosing the sampler to match the objective is negligible.

623 5.6. Posterior uncertainty as a stopping signal

624 The adaptive stopping rule discussed in Section 6.2 is viable only if the
 625 surrogate’s own uncertainty is informative about its actual error. Fig. 12
 626 compares the Gaussian-process posterior standard deviation σ with the ab-
 627 solute reconstruction error $|\Delta\eta|$ for the ALC design at $n = 12$ (seed 0). The
 628 two fields share the same spatial structure: both attain their largest values in
 629 the sparsely sampled low-discharge, low-head corner of the envelope and both
 630 are small throughout the well-covered interior. The agreement is quantitative
 631 as well as visual, with a Pearson correlation of $r = 0.70$ between σ and $|\Delta\eta|$
 632 at the 72 measured points ($p < 10^{-11}$). The posterior uncertainty is there-
 633 fore a usable running proxy for where, and by how much, the reconstruction

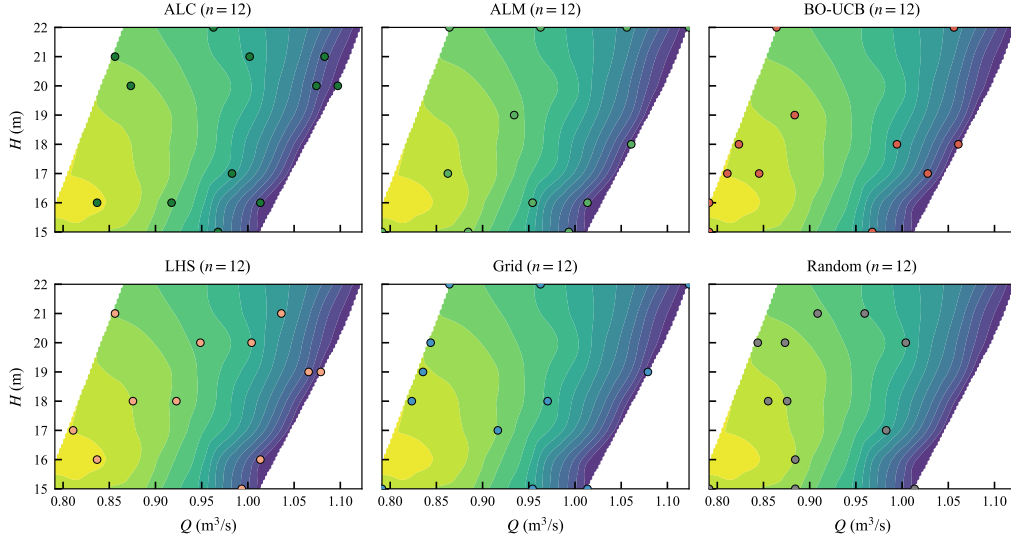


Figure 8: Sample placement at $n = 12$ for all six policies over the ground-truth efficiency contour: the active learners and space-filling designs cover the envelope, whereas BO-UCB concentrates on the high-efficiency ridge.

remains untrustworthy, which is precisely the property an uncertainty-based stopping criterion relies on.

5.7. Practical guidance: choosing the budget and the sampler

The preceding results translate into a compact recipe that an engineer can apply before a campaign begins, summarized in Table 4. The decision has two inputs: the deliverable (the full hill chart or only the BEP) and the budget the campaign can afford. For the common deliverable—the full hill chart—the sweet spot is $n = 10$ to 12 experiments, an 86% to 83% reduction from the 72-point campaign. This is the smallest budget at which the active learners reach engineering-grade accuracy ($R^2 \geq 0.99$, $\text{MAPE} < 0.5\%$) while remaining clear of the $n < 8$ regime in which the Gaussian process is fitted on too few points to be reliable (Section 5.5); we recommend $n = 12$ as the conservative default and $n = 10$ when experiments are especially scarce. Below eight experiments, where a model-based policy cannot yet be trusted, a space-filling Latin hypercube design is the safer choice and still recovers the surface to $R^2 \approx 0.97$ at an 89% to 92% reduction. If the deliverable is only the BEP, 14 experiments (an 81% reduction) with BO-UCB are sufficient to locate the exact optimum node. Because ALM and ALC are statistically

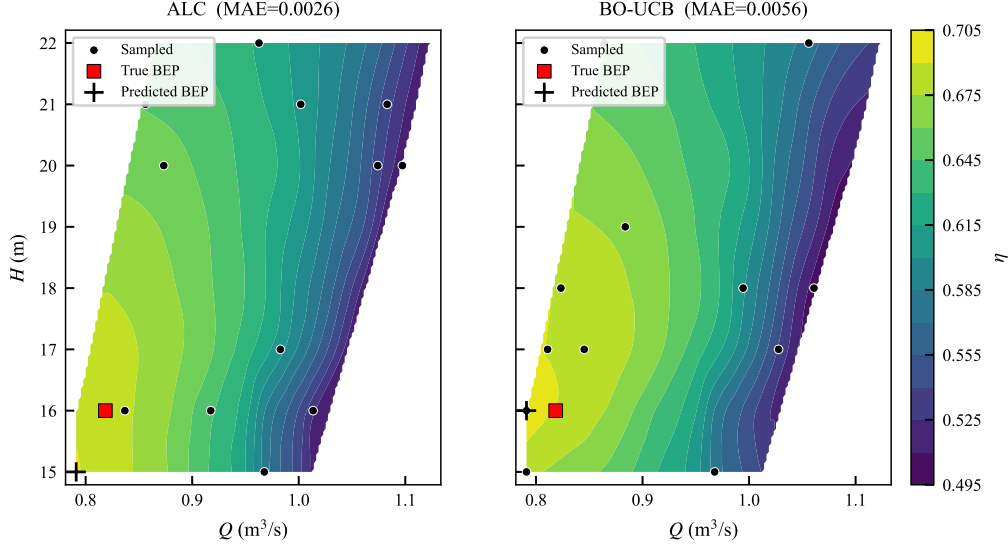


Figure 9: The two goals in one frame: ALC (left) and BO-UCB (right) at the same budget $n = 12$. BO-UCB pins the BEP but distorts the map; ALC recovers the map but locates the BEP less precisely. This contrast is the thesis of the paper: the acquisition function must match the deliverable.

indistinguishable for reconstruction (Section 5.4), we recommend ALM as the default reconstruction sampler on the grounds of lower computational cost. Two caveats travel with the table: the budgets are calibrated on a smooth, unimodal low-head propeller hill chart and should be revisited for machines with more structured efficiency surfaces, and the very-low-budget recommendation assumes the practitioner cannot afford the space-filling initialization that would stabilize a model-based policy.

6. Discussion and Conclusion

6.1. Interpreting the exploitation advantage

The central finding of this study is that the policy ranked worst by global surface-reconstruction metrics is the best for locating the best efficiency point, and that this is not a contradiction but a direct and predictable consequence of how an exploitation-driven adaptive policy allocates a limited budget. Bayesian optimization with an upper-confidence-bound criterion concentrates experiments near the apparent optimum and along the

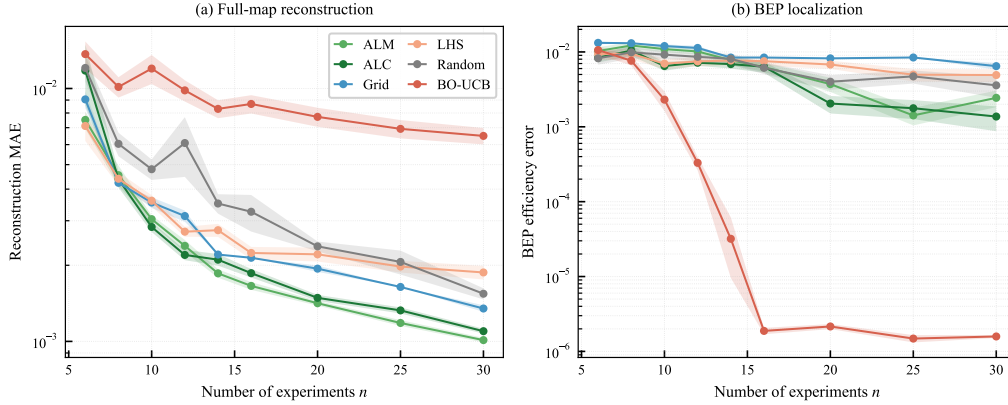


Figure 10: Learning curves over the budget sweep (mean $\pm$ standard error, 20 seeds). (a) Global reconstruction versus n : the active learners (ALC, ALM) dominate and BO-UCB is worst. (b) BEP efficiency error versus n : BO-UCB dominates, reaching the level of numerical noise by $n = 14$, while the others plateau. The inversion between the panels is the core result.

high-efficiency ridge, and declines to sample the low-efficiency, high-discharge corner (Fig. 8). That corner is therefore reconstructed poorly, which inflates global error statistics, while the peak region is resolved essentially exactly, which is what the localization metrics reward: from 14 experiments the policy identifies the exact BEP node on every seed. The active learners exhibit the mirror-image behavior: by directing their budget at the surrogate’s remaining uncertainty they reconstruct the whole surface accurately, including the parts of it that no operator cares about, but they never place enough experiments at the peak to resolve the optimum sharply, and their localization error plateaus near 10^{-2} . The methodological lesson is that the success criterion must match the decision objective. A study that adopts the global coefficient of determination as its headline metric will rank the active learners and space-filling designs first and dismiss the adaptive optimizer, and will thereby recommend a test plan optimized for a goal—uniform global fidelity—that a BEP-directed engineer did not actually have. When the goal is BEP identification, the appropriate criteria are localization accuracy and reliability, and under those criteria the optimization policy is decisively preferred. This realignment of metric to objective, rather than any single numerical result, is the contribution most likely to transfer to other turbine performance-testing campaigns.

Table 3: Wilcoxon signed-rank tests at $n = 12$ (20 seeds, paired by seed). Top: reconstruction MAE, reference ALC. Bottom: BEP efficiency error, reference BO-UCB. $p < 0.05$ is significant.

Comparison	Metric	p -value	Significant?
ALC vs. ALM	MAE	0.09	no (equivalent)
ALC vs. LHS	MAE	0.009	yes
ALC vs. Grid	MAE	1.3×10^{-4}	yes
ALC vs. Random	MAE	1.3×10^{-5}	yes
ALC vs. BO-UCB	MAE	9.5×10^{-7}	yes
BO-UCB vs. ALC	$\Delta\eta$	9.5×10^{-7}	yes
BO-UCB vs. ALM	$\Delta\eta$	9.5×10^{-7}	yes
BO-UCB vs. LHS	$\Delta\eta$	4.8×10^{-6}	yes
BO-UCB vs. Grid	$\Delta\eta$	9.5×10^{-7}	yes
BO-UCB vs. Random	$\Delta\eta$	9.5×10^{-7}	yes

6.2. Decision-support implications

Framing performance testing as sequential decision-making yields a practical benefit beyond the choice of sampling policy. Because the Gaussian process surrogate reports a posterior uncertainty that demonstrably tracks the actual reconstruction error (Section 5.6, Fig. 12), it provides the engineer with a running estimate of confidence in the current map and, in particular, in the estimated BEP. This supports an adaptive stopping rule: testing can continue until the posterior uncertainty in the high-efficiency region, or the acquisition value available from a further experiment, falls below a threshold the engineer is willing to accept, rather than until a pre-committed number of points has been measured. The budget sweep suggests that such a rule would have terminated testing at roughly 14 experiments for a BEP-directed campaign on the present machine, and at 10 to 12 experiments for hill-chart reconstruction, against the 72 of the full campaign. The non-adaptive designs cannot support such a rule, because they do not use the observed efficiencies to guide sampling: their sample locations are fixed before any data are seen and cannot respond to where the surrogate remains most uncertain. In a setting where each experiment is genuinely expensive, the ability to decide when to stop on a principled basis is as valuable as the ability to decide where to test next, and both follow from the decision-theoretic framing adopted here.

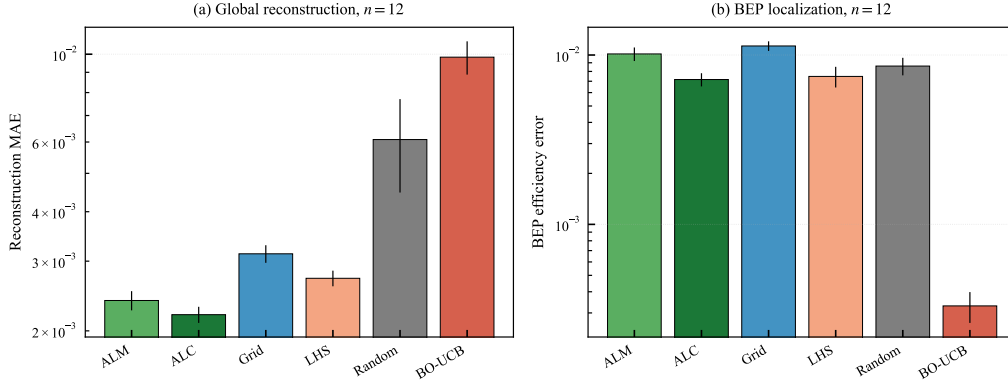


Figure 11: Method ranking at $n = 12$ with standard-error bars. (a) Reconstruction: active learners best. (b) BEP localization: BO-UCB best. The ranking inverts between the two objectives.

6.3. Limitations

Several limitations bound the conclusions of this study. First, the ground-truth BEP head coincides exactly with an integer head node of the test matrix. This coincidence favors any policy that samples that node and contributes to the exact BEP recovery reported for the Bayesian policy; a perturbation analysis in which the BEP is defined on a continuous interpolant, or in which the test grid is offset, would establish how much of the localization advantage survives when the optimum does not lie on a node. Second, the near-zero localization error of the Bayesian policy reflects direct measurement of the peak region rather than superior out-of-sample interpolation there: the policy spends its budget inside the high-efficiency envelope, and its advantage should be understood accordingly. Third, all sequential policies are initialized at random, and below about eight experiments the Gaussian process is fitted on too few points for the model-based acquisitions to be reliable (Section 5.5); a structured initialization, such as the corners-and-center core of a central composite design, might improve this very-low-budget behavior and remains to be compared systematically. Fourth, the study concerns one machine and one measured dataset, so the generality of the finding across other propeller turbines, other turbine types, and other operating envelopes is not established. None of these limitations undermines the qualitative finding, but each constrains how strongly it can be stated.

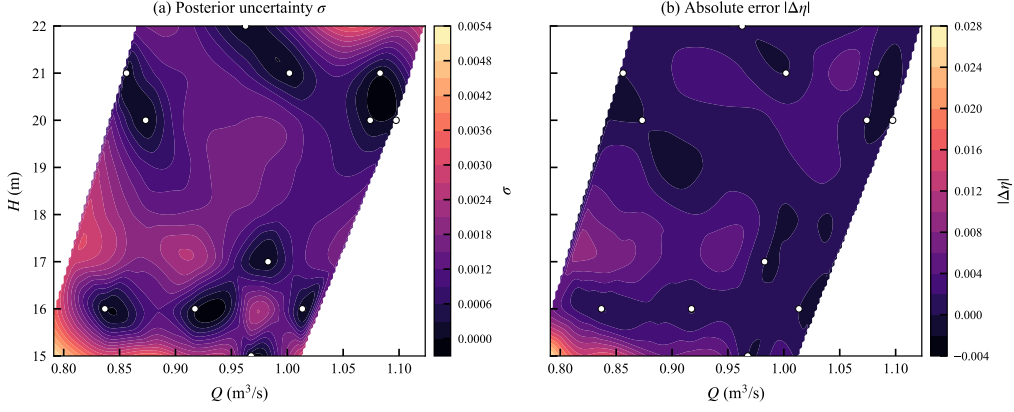


Figure 12: Gaussian-process posterior uncertainty σ (a) and absolute reconstruction error $|\Delta\eta|$ (b) for the ALC design at $n = 12$ (seed 0), with the 12 sampled points marked. The two fields are strongly correlated (Pearson $r = 0.70$ at the 72 measured points), supporting the use of posterior uncertainty as a stopping signal.

6.4. Conclusion and future work

This paper has formulated the construction of a small-hydro propeller-turbine performance map from physical experiments as a sequential, budget-constrained decision problem, and has benchmarked six sampling policies—Bayesian optimization, two Gaussian-process active-learning policies, and random, grid, and Latin hypercube designs—on a measured ground-truth dataset of 72 points, with the best efficiency point as an explicit decision target. The study finds that the two engineering objectives select opposite winners. For reconstructing the full hill chart, the active learners dominate, reaching $R^2 \geq 0.99$ from as few as 10 experiments and $R^2 \approx 0.996$ from 12 (an 83.3% reduction), while the optimization policy is the weakest map-builder. For locating the BEP the ranking inverts: Bayesian optimization identifies the exact BEP node on every seed from 14 experiments (an 81% reduction), while every other policy plateaus near $\Delta\eta \approx 10^{-2}$. The practical implication is that turbine performance testing should be planned as a targeted decision process whose sampling policy—and whose success metric—matches the deliverable: active learning when the deliverable is the hill chart, Bayesian optimization when it is the BEP, and a space-filling design as the robust fallback at the very smallest budgets. For the practitioner these findings reduce to a concrete budget recipe (Table 4): the hill chart is recovered at the sweet spot of 10 to 12 experiments (an 83% to 86% reduction) and the BEP

Table 4: Practical guidance for a reduced test campaign on a smooth low-head propeller hill chart. Reduction is relative to the 72-point reference campaign; accuracy figures are the mean over 20 seeds.

Deliverable	Recommended Budget n		Reduction	Expected accuracy
	sampler			
Full hill chart (default)	ALM (or ALC)	12	83%	$R^2 \approx 0.996$, MAPE 0.38%
Full hill chart (aggressive)	ALC	10	86%	$R^2 \approx 0.99$, MAPE < 0.5%
Full hill chart (extreme, model-free)	LHS	6–8	89–92%	$R^2 \approx 0.97$
Best efficiency point only	BO-UCB	14	81%	exact BEP node, $\Delta\eta \rightarrow 0$

at 14 experiments (an 81% reduction), with Latin hypercube sampling as the safe choice below eight experiments. Although demonstrated on a single small-hydro propeller turbine, the sequential-decision framing is not specific to this machine: objective-directed adaptive sampling has been applied to the optimization of other small-scale energy-conversion devices [33], and the present results suggest that comparable budget savings should transfer to any expensive performance-mapping campaign in which the deliverable is well defined.

Future work should test the sensitivity of the result to the placement of the BEP relative to the test grid, compare structured initialization schemes against the random initialization used here, examine the robustness of the ranking to the choice of GP kernel, and extend the comparison to a second measured dataset, other turbine types, and other operating envelopes. Incorporating additional decision objectives—such as the joint consideration of efficiency and cavitation margin—within the same sequential-decision framework, and integrating a principled stopping rule based on posterior uncertainty into a live test campaign, are promising directions for reducing the cost of turbine performance characterization in practice.

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775 **Conflict of Interest**

776 The authors declare that they have no known competing financial inter-
777 ests or personal relationships that could have appeared to influence the work
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779 **Declaration of generative AI in scientific writing**

780 During the preparation of this work, the authors used Claude (Anthropic)
781 as a coding assistant to implement the analysis pipeline, generate the figures,
782 and refine the language of the manuscript. The research idea, problem formu-
783 lation, methodology, and all interpretations and conclusions are the authors’
784 own. Every AI-assisted output was reviewed, verified, and edited by the au-
785 thors, who take full responsibility for the content and integrity of the work.

786 **CRedit Author Contribution Statement**

787 **Siwakorn Suprasertchai:** Conceptualization, Methodology, Software,
788 Validation, Visualization, Project administration. **Kornpaphop Ruttanaw-**
789 **ijit:** Data Curation, Writing – Original Draft. **Yodchai Tiaple:** Conceptu-
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792 **References**

- 793 [1] D. Zhou, Z. D. Deng, Ultra-low-head hydroelectric technology: A review,
794 Renewable and Sustainable Energy Reviews 78 (2017) 23–30. doi:10.
795 1016/j.rser.2017.04.086.
- 796 [2] S. Phitaksurachai, R. Pan-Aram, N. Sritrakul, Y. Tiaple, Performance
797 testing of low head small hydro power development in thailand, Energy
798 Procedia 138 (2017) 1140–1146. doi:10.1016/j.egypro.2017.10.221.

- 799 [3] International Electrotechnical Commission, IEC 60193:2019 hydraulic
800 turbines, storage pumps and pump-turbines — model acceptance tests,
801 Tech. rep., International Electrotechnical Commission, Geneva, Switzer-
802 land (2019).
- 803 [4] Z. Masood, S. Khan, L. Qian, Machine learning-based surrogate model
804 for accelerating simulation-driven optimisation of hydropower kaplan
805 turbine, *Renewable Energy* 173 (2021) 827–848. doi:10.1016/j.
806 *renene*.2021.04.005.
- 807 [5] F. Besni, F. Büyüksolak, E. Ayli, K. Celebioglu, S. Aradag, Y. Tas-
808 cioglu, Machine learning-based efficiency prediction of francis type hy-
809 draulic turbines through comprehensive performance testing, *Proceed-*
810 *ings of the Institution of Mechanical Engineers, Part A: Journal of Power*
811 *and Energy* 239 (2025) 755–775. doi:10.1177/09576509251352663.
- 812 [6] J. Betancour, F. Romero-Menco, L. Velásquez, A. Rubio-Clemente,
813 E. Chica, Design and optimization of a runner for a gravitational vortex
814 turbine using the response surface methodology and experimental tests,
815 *Renewable Energy* 210 (2023) 306–320. doi:10.1016/j.*renene*.2023.
816 04.045.
- 817 [7] A. Telikani, M. Rossi, N. Khajehali, M. Renzi, Pumps-as-turbines’ (pats)
818 performance prediction improvement using evolutionary artificial neural
819 networks, *Applied Energy* 330 (2023) 120316. doi:10.1016/j.
820 *apenergy*.2022.120316.
- 821 [8] W. Yu, P. Zhou, Z. Miao, H. Zhao, J. Mou, W. Zhou, Energy per-
822 formance prediction of pump as turbine (pat) based on piwoa-bp neu-
823 ral network, *Renewable Energy* 222 (2024) 119873. doi:10.1016/j.
824 *renene*.2023.119873.
- 825 [9] S. M. Malakouti, Use machine learning algorithms to predict turbine
826 power generation to replace renewable energy with fossil fuels, *En-*
827 *ergy Exploration & Exploitation* 41 (2023) 836–857. doi:10.1177/
828 01445987221138135.
- 829 [10] Z. Zou, P. Xu, Y. Chen, L. Yao, C. Fu, Application of artificial
830 intelligence in turbomachinery aerodynamics: progresses and chal-
831 lenges, *Artificial Intelligence Review* 57 (2024) 222. doi:10.1007/
832 s10462-024-10867-3.

- [11] S. S. Talebi, A. Madadi, A. M. Tousi, M. Kiaee, Micro gas turbine fault detection and isolation with a combination of artificial neural network and off-design performance analysis, *Engineering Applications of Artificial Intelligence* 113 (2022) 104900. doi:10.1016/j.engappai.2022.104900.
- [12] G. E. P. Box, K. B. Wilson, On the experimental attainment of optimum conditions, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 13 (1951) 1–38. doi:10.1111/j.2517-6161.1951.tb00067.x.
- [13] M. D. McKay, R. J. Beckman, W. J. Conover, Comparison of three methods for selecting values of input variables in the analysis of output from a computer code, *Technometrics* 21 (1979) 239–245. doi:10.1080/00401706.1979.10489755.
- [14] J. Taghinezhad, S. Abdoli, V. Silva, S. Sheidaei, R. Alimardani, E. Mahmoodi, Computational fluid dynamic and response surface methodology coupling: A new method for optimization of the duct to be used in ducted wind turbines, *Heliyon* 9 (2023) e17057. doi:10.1016/j.heliyon.2023.e17057.
- [15] P. Borisut, A. Nuchitprasittichai, Adaptive latin hypercube sampling for a surrogate-based optimization with artificial neural network, *Processes* 11 (2023) 3232. doi:10.3390/pr11113232.
- [16] D. R. Jones, M. Schonlau, W. J. Welch, Efficient global optimization of expensive black-box functions, *Journal of Global Optimization* 13 (1998) 455–492. doi:10.1023/A:1008306431147.
- [17] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, N. de Freitas, Taking the human out of the loop: A review of bayesian optimization, *Proceedings of the IEEE* 104 (2016) 148–175. doi:10.1109/JPROC.2015.2494218.
- [18] C. E. Rasmussen, C. K. I. Williams, *Gaussian Processes for Machine Learning*, The MIT Press, 2006. doi:10.7551/mitpress/3206.001.0001.
- [19] N. Srinivas, A. Krause, S. M. Kakade, M. W. Seeger, Information-theoretic regret bounds for gaussian process optimization in the bandit

- 866 setting, IEEE Transactions on Information Theory 58 (2012) 3250–3265.
867 doi:10.1109/TIT.2011.2182033.
- 868 [20] J. Snoek, H. Larochelle, R. P. Adams, Practical bayesian optimization
869 of machine learning algorithms, in: Advances in Neural Information
870 Processing Systems 25 (NeurIPS 2012), Curran Associates, Red Hook,
871 NY, 2012, pp. 2951–2959.
- 872 [21] S. Greenhill, S. Rana, S. Gupta, P. Vellanki, S. Venkatesh, Bayesian
873 optimization for adaptive experimental design: A review, IEEE Access
874 8 (2020) 13937–13948. doi:10.1109/ACCESS.2020.2966228.
- 875 [22] Q. Liang, A. E. Gongora, Z. Ren, A. Tiihonen, Z. Liu, S. Sun, J. R.
876 Deneault, D. Bash, F. Mekki-Berrada, S. A. Khan, K. Hippalgaonkar,
877 B. Maruyama, K. A. Brown, J. Fisher III, T. Buonassisi, Benchmarking
878 the performance of bayesian optimization across multiple experimental
879 materials science domains, npj Computational Materials 7 (2021) 188.
880 doi:10.1038/s41524-021-00656-9.
- 881 [23] J. Chen, C. Liu, Efficient one-dimensional turbomachinery design
882 method based on transfer learning and bayesian optimization, SN Ap-
883 plied Sciences 4 (2022) 255. doi:10.1007/s42452-022-05132-7.
- 884 [24] R.-L. Liu, Q. Zhao, X.-J. He, X.-Y. Yuan, W.-T. Wu, M.-Y. Wu,
885 Airfoil optimization based on multi-objective bayesian, Journal of Me-
886 chanical Science and Technology 36 (2022) 5561–5573. doi:10.1007/
887 s12206-022-1020-y.
- 888 [25] S. Lim, A. Garbo, P. Bekemeyer, C. Appel, R. Ewert, J. Delfs, High-
889 fidelity aerodynamic and aeroacoustic multi-objective bayesian opti-
890 mization, in: AIAA AVIATION 2022 Forum, American Institute of
891 Aeronautics and Astronautics, Chicago, IL, 2022. doi:10.2514/6.
892 2022-3354.
- 893 [26] L. Pretsch, I. Arsenyev, N. Bartoli, F. Duddeck, Bayesian optimization
894 of cooperative components for multi-stage aero-structural compressor
895 blade design, Structural and Multidisciplinary Optimization 68 (2025)
896 84. doi:10.1007/s00158-025-03998-w.
- 897 [27] Y. Guo, P. Nath, S. Mahadevan, P. Witherell, Active learning for
898 adaptive surrogate model improvement in high-dimensional problems,

- 899 Structural and Multidisciplinary Optimization 67 (2024) 122. doi:
900 10.1007/s00158-024-03816-9.
- 901 [28] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion,
902 O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vander-
903 plas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, E. Duchesnay,
904 Scikit-learn: Machine learning in python, Journal of Machine Learning
905 Research 12 (2011) 2825–2830.
- 906 [29] P. S. Palar, L. R. Zuhail, K. Shimoyama, Gaussian process surrogate
907 model with composite kernel learning for engineering design, AIAA Jour-
908 nal 58 (2020) 1864–1880. doi:10.2514/1.J058807.
- 909 [30] D. J. C. MacKay, Information-based objective functions for active data
910 selection, Neural Computation 4 (1992) 590–604. doi:10.1162/neco.
911 1992.4.4.590.
- 912 [31] D. A. Cohn, Z. Ghahramani, M. I. Jordan, Active learning with statisti-
913 cal models, Journal of Artificial Intelligence Research 4 (1996) 129–145.
914 doi:10.1613/jair.295.
- 915 [32] J. Sacks, W. J. Welch, T. J. Mitchell, H. P. Wynn, Design and analysis
916 of computer experiments, Statistical Science 4 (1989) 409–423. doi:
917 10.1214/ss/1177012413.
- 918 [33] Z. Sefidgar, A. Ashrafizadeh, A. Arabkoohsar, Techno-economic analysis
919 and multi-objective optimization of cross-flow wind turbines for smart
920 building energy systems, Global Challenges 7 (2023) 2200203. doi:
921 10.1002/gch2.202200203.